

PSLE MATHEMATICS PRACTICE PAPER 9

Mixed Problems & Applications

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. If 5 pens cost \$2.50, how much do 8 pens cost? (2 marks)

2. A recipe needs 2 cups of flour for 12 cookies. How many cups for 36 cookies? (2 marks)

3. In a class, the ratio of boys to girls is 3:2. If there are 15 boys, how many girls? (2 marks)

4. A discount of 15% is given on an item costing \$80. Find the sale price. (2 marks)

5. If 4 workers can complete a job in 6 days, how many days for 2 workers? (2 marks)

6. A bottle contains 750 ml. How many liters is this? (2 marks)

7. The sum of two consecutive numbers is 25. Find the numbers. (2 marks)

8. Convert 2 hours 45 minutes to minutes. (2 marks)

SECTION B: Long Answer Questions (30 marks)

9. Sarah spends $\frac{1}{4}$ of her money on food, $\frac{1}{3}$ on transport, and saves the rest. If she saves \$45, how much money did she have? (4 marks)

10. A map scale is 1:50000. If two towns are 3 cm apart on the map, what is the actual distance in km? (3 marks)

11. A shopkeeper bought 100 items at \$5 each and sold them at \$8 each. Find his profit. (3 marks)

12. If 3 people share \$120 in the ratio 1:2:3, how much does each get? (4 marks)

13. A water tank is $\frac{3}{4}$ full. It contains 900 liters. What is its capacity? (2 marks)

14. Simplify: (a) $24 \div 6 + 8 \times 2$, (b) $12 + 3^2 - 1$ (4 marks)

15. A car uses 1 liter of petrol for every 12 km. How much petrol is needed for a 360 km journey? (3 marks)

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SOLUTIONS

SECTION A SOLUTIONS

Q1: Cost of pens

Cost per pen = $\$2.50 \div 5 = \0.50 , 8 pens = $8 \times \$0.50 = \4

Q2: Recipe scaling

2 cups for 12, for 36: $(36 \div 12) \times 2 = 6$ cups

Q3: Boys to girls ratio

3:2 ratio, 15 boys means $15 \div 3 = 5$ per unit, Girls = $2 \times 5 = 10$

Q4: Discount

15% of \$80 = \$12, Sale price = $\$80 - \$12 = \$68$

Q5: Inverse proportion

4 workers $\times$ 6 days = 2 workers $\times$ x days, $x = 12$ day

Q6: Milliliters to liters

750 ml = $750 \div 1000 = 0.75$ liters

Q7: Consecutive numbers

Let n and n+1, $n + n + 1 = 25$, $2n + 1 = 24$, $n = 12$. Numbers: 12 and 13

Q8: Convert time

2 hours 45 min = $(2 \times 60) + 45 = 165$ minutes

SECTION B SOLUTIONS

Q9: Money problem

$\frac{1}{4} + \frac{1}{3} = \frac{3}{12} + \frac{4}{12} = \frac{7}{12}$ spent, $\frac{5}{12}$ saved = \$45, Total = $\$45 \times \frac{12}{5} = \108

Q10: Map scale

1 cm = 50000, 3 cm = 150000 cm = 1.5 km

Q11: Profit

Cost = $100 \times \$5 = \500 , Revenue = $100 \times \$8 = \800 , Profit = \$300

Q12: Share ratio 1:2:3

Total parts = 6, Each: $\$120 \div 6 = \20 , \$40, \$60

Q13: Tank capacity

$\frac{3}{4}$ capacity = 900, Capacity = $900 \times \frac{4}{3} = 1200$ liters

Q14: Order of operations

(a) $24 \div 6 + 8 \times 2 = 4 + 16 = 20$, (b) $12 + 9 - 5 = 16$

Q15: Petrol needed

$360 \text{ km} \div 12 \text{ km per liter} = 30$ liters

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